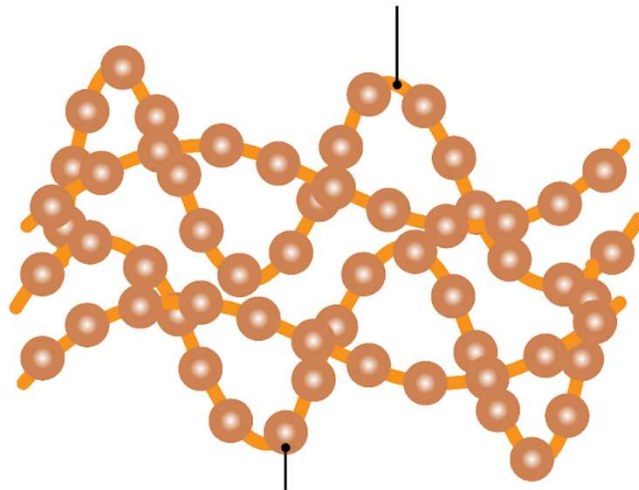


	Thermoplastic Polymer	Thermosetting Polymer
Definition	Polymers that soften on heating and harden on cooling . This dcan be repeated many times.	Polymers that permanently harden on heating due to extensive cross-linking. Cannot be remelted or reshaped once set.
Structure	Have linear or slightly branched polymer chains. Weak intermolecular forces.	Have highly cross-linked , three-dimensional network structure. Strong covalent bonds between chains.
Behavior on Heating	Melt and become soft. Can be reshaped multiple times.	Do not melt ; instead, they decompose or char when heated after curing.
Mechanical Properties	Generally softer, flexible , and less brittle . Lower melting points	Hard, rigid , and brittle . High thermal and chemical resistance.
Recyclability	Recyclable (can be remelted).	Not recyclable by melting (due to permanent cross-links).
Examples	Polyethylene (PE), Polypropylene (PP),PVC, Polystyrene (PS),Nylon	Bakelite,Epoxy resin,Melamine formaldehyde,Urea formaldehyde,Phenol formaldehyde
Uses	Packaging, bottles, toys, pipes, insulating material.	Electrical switches, circuit boards, adhesives, kitchenware, automotive parts.

Thermoplastic

Strong link into polymer chains



Monomer

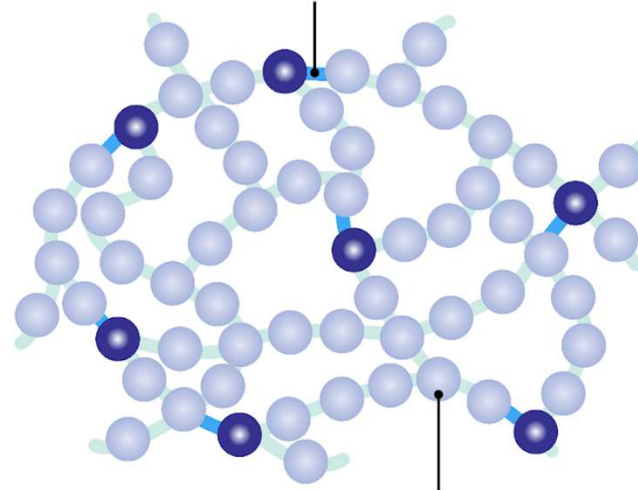
Weak intermolecular forces
between polymer chains

No cross-links between chains

Softens when heated

Thermosetting

Strong cross-link bond



Monomer

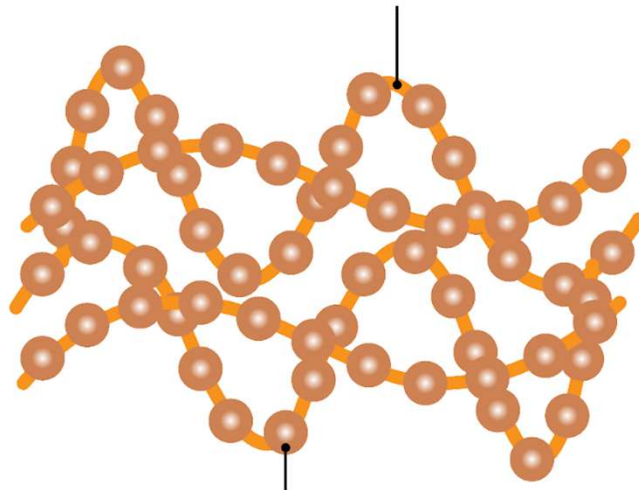
Strong covalent bonds
between polymer chains

Remains hard when heated

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Thermoplastic

Strong link into polymer chains



Monomer

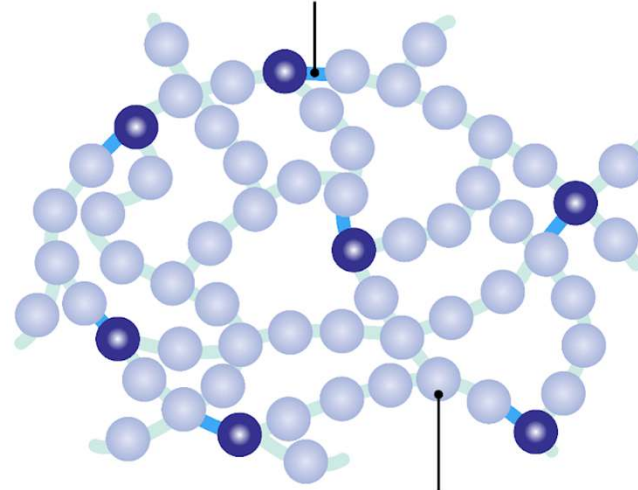
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Monomer

Strong covalent bonds
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